

SIMATS ENGINEERING

ASSESSMENT TOOL 2: Research Review & Analysis

COURSE NAME: ARTIFICIAL INTELLIGENCE COURSE CODE: MLA01
AND EXPERT SYSTEMS

COURSE OUTCOME COVERED

CO4: *Develop intelligent software agents using suitable agent architectures and learning techniques.*(BTL 3)

Assessment Tool Weightage: 30%

Total Marks: 100

Time: 90 Mins

No. of Questions: 1

Annexure A Research Review & Analysis

Code of Conduct:

I SHAM S 192325076 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criterion	Weightage	Marks Obtained
1. Problem Analysis and Domain Understanding	10%	
2. Literature Search and Source Selection	10%	
3. Literature Review and Synthesis	15%	
4. Comparative Analysis of AI Techniques	10%	
5. Critical Analysis of Research Findings	10%	
6. Research Gap Identification	10%	
7. Proposed Research Direction	10%	
8. Research Methodology / Analysis Framework	5%	
9. Experimental / Evidence-Based Validation	10%	
10. Result Analysis and Interpretation	5%	

11. Research Paper Quality and Referencing	10%	
12. Technical Presentation and Research Defense	5%	
Total	100%	

Signature of the Faculty

Total Marks Awarded

ANSWER ANY ONE QUESTION

S.No.	Research Review & Analysis Question	Primary Topic
1	Decision Tree Learning: Review recent research on ID3, C4.5, and CART. Compare their attribute-selection methods, pruning strategies, interpretability, computational complexity, and classification performance. Identify a research gap and suggest a possible improvement.	Decision Trees
2	Neural Networks vs Statistical Learning: Review and compare research on neural-network and statistical-learning approaches for prediction and classification. Analyze their performance, data requirements, generalization, interpretability, computational cost, and suitability for different applications.	Statistical Learning & Neural Networks
3	Kernel Methods: Analyze research on SVM and kernel methods for non-linear classification. Compare linear, polynomial, and RBF kernels and evaluate their advantages and limitations compared with neural-network-based approaches.	Kernel Methods
4	Reinforcement Learning: Review research on Q-Learning, SARSA, and Deep Q-Networks. Compare their learning mechanisms, exploration strategies, convergence, computational requirements, and real-world applications. Identify challenges and research gaps.	Reinforcement Learning

5	Explainable and Generalizable AI: Review research on Inductive Learning and Explanation-Based Learning and analyze how these approaches contribute to generalization and explainability in AI. Identify current limitations and propose future research directions.	Inductive & Explanation-Based Learning
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1. Abstract

Decision Tree learning is one of the most widely studied approaches for supervised classification because it represents learned decisions in a human-interpretable tree structure. Among the classical algorithms, **ID3, C4.5 and CART** are particularly important because they represent different approaches to attribute selection, tree construction and pruning.

This research review examines these three algorithms with emphasis on their **attribute-selection criteria, pruning strategies, handling of data characteristics, interpretability, computational complexity, scalability and classification performance**. ID3 uses Information Gain, C4.5 extends ID3 using Gain Ratio and introduces additional capabilities such as handling continuous attributes and post-pruning, while CART generally constructs binary trees using criteria such as the Gini Index and applies cost-complexity pruning.

Research comparing these algorithms shows that no single method is consistently optimal across all datasets. Performance can depend on dataset size, attribute types, noise, missing values and the evaluation criterion. Recent comparative work has also examined tree depth, leaf-node count, construction time and classification accuracy, indicating that computational efficiency and predictive performance can differ substantially between algorithms.

The review identifies a research opportunity in developing **adaptive decision-tree learning methods that select splitting and pruning strategies according to dataset characteristics**, rather than relying on one fixed criterion. A possible direction is a hybrid adaptive tree that dynamically selects among entropy-based, gain-ratio and Gini-based splitting while controlling complexity through validation-driven pruning.

2. Keywords

Decision Tree, ID3, C4.5, CART, Attribute Selection, Pruning

3. Introduction

Decision Trees are supervised machine-learning models that classify observations by recursively partitioning a dataset according to attribute-based decision rules. Their major advantage is that the resulting model can be represented as a tree, allowing users to follow a sequence of decisions from the root to a leaf.

The development of modern Decision Tree learning includes several influential algorithms. **ID3**, introduced by J. Ross Quinlan, established an important approach to inductive decision-tree construction using information-theoretic attribute selection. Quinlan's original work describes ID3 and discusses issues involving noisy and incomplete information.

C4.5 was subsequently developed as an extension of the ID3 approach. Quinlan's 1993 C4.5 system supports mixtures of nominal and numeric attributes and incorporates mechanisms for issues such as missing data and overfitting. It can produce decision trees as well as rule-based representations.

CART, introduced by Breiman, Friedman, Olshen and Stone, provides a broader framework for Classification and Regression Trees. The methodology covers tree construction, splitting rules and pruning, including minimal cost-complexity pruning.

These algorithms remain important because Decision Trees provide a useful balance between **predictive modelling and interpretability**. However, the algorithms differ significantly in their splitting criteria, tree structures, pruning methods and computational behaviour.

Therefore, this review investigates the differences among ID3, C4.5 and CART and examines whether one algorithm can be considered universally superior.

4. Research Problem and Objectives

4.1 Research Problem

Although ID3, C4.5 and CART all generate Decision Trees, they use different strategies to select attributes, construct trees and control overfitting.

The central research problem is:

How do ID3, C4.5 and CART differ in attribute selection, pruning, interpretability, computational complexity and classification performance, and under what dataset conditions is each algorithm most appropriate?

4.2 Objectives

The objectives of this research review are:

1. To study the principles of ID3, C4.5 and CART.
 2. To compare their attribute-selection methods.
 3. To analyse their pruning strategies.
 4. To compare their interpretability.
 5. To examine computational complexity and scalability.
 6. To analyse reported classification performance.
 7. To identify strengths and limitations of each algorithm.
 8. To identify a research gap in existing Decision Tree approaches.
 9. To propose a technically feasible improvement.
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5. Literature Review

5.1 ID3

ID3 was introduced by **J. Ross Quinlan in 1986**. The algorithm was designed for inductive construction of decision trees from training examples. Its attribute-selection process is based on **Information Gain**, which uses entropy to measure the uncertainty associated with a dataset.

The general principle is:

Select the attribute that provides the largest reduction in entropy.

ID3 is relatively simple and produces interpretable decision trees. However, the original approach has limitations. Research comparing classical tree algorithms notes that ID3 does not originally provide the same capabilities as later algorithms for numeric attributes, missing values and pruning.

Main characteristics

- Attribute selection: **Information Gain**
 - Tree structure: Generally multiway
 - Original pruning: **No dedicated pruning mechanism**
 - Numerical attributes: Limited in original formulation
 - Missing values: Limited in original formulation
 - Interpretability: High
 - Computational simplicity: Relatively high
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5.2 C4.5

C4.5 was developed by Quinlan as an extension of ID3.

One of its important improvements is the use of **Gain Ratio** rather than raw Information Gain. This helps reduce the tendency of Information Gain to favour attributes having many distinct values.

C4.5 also supports **continuous/numeric attributes**, missing values and post-pruning. The C4.5 system was designed to generate understandable models while maintaining classification capability.

C4.5 applies **error-based pruning** after tree growth. Research descriptions of C4.5 identify post-pruning as an important part of the algorithm.

Main characteristics

- Attribute selection: **Gain Ratio**
 - Tree structure: Multiway
 - Pruning: **Error-based post-pruning**
 - Numerical attributes: Supported
 - Missing values: Supported
 - Interpretability: High
 - Complexity: Higher than basic ID3 because of additional processing
-

5.3 CART

CART stands for **Classification and Regression Trees** and was developed by Breiman, Friedman, Olshen and Stone.

Unlike ID3 and C4.5, CART generally constructs **binary trees**, meaning each internal node produces two branches.

For classification, CART commonly uses the **Gini Index** as its splitting criterion. The CART framework also includes mechanisms for dealing with classification costs and prior probabilities.

A major characteristic of CART is **cost-complexity pruning**, which attempts to balance tree accuracy against tree complexity. The original CART work explicitly covers minimal cost-complexity pruning and methods for producing appropriately sized trees.

Main characteristics

- Attribute selection: **Gini Index** commonly used
 - Tree structure: **Binary**
 - Pruning: **Cost-complexity pruning**
 - Numerical attributes: Supported
 - Missing values: Can be handled through CART mechanisms
 - Interpretability: High
 - Computational cost: Can increase with continuous-feature split evaluation
-

6. Comparative Analysis

6.1 Overall Comparison

Feature	ID3	C4.5	CART
Developer	J.R. Quinlan	J.R. Quinlan	Breiman et al.
Main criterion	Information Gain	Gain Ratio	Gini Index
Tree structure	Multiway	Multiway	Binary
Pruning	Original ID3: No	Error-based pruning	Cost-complexity pruning
Numeric attributes	Limited originally	Yes	Yes
Missing values	Limited originally	Yes	Supported
Interpretability	High	High	High
Classification	Yes	Yes	Yes
Regression	No	Primarily classification	Yes
Overfitting control	Weak in original form	Stronger	Strong
Scalability	Relatively simple	More processing	Efficient but split search can be costly
Main strength	Simplicity	Flexible improvement over ID3	Robust tree framework

The fundamental differences are supported by comparative research describing ID3, C4.5 and CART in terms of splitting criteria and pruning mechanisms.

6.2 Attribute Selection

ID3 — Information Gain

ID3 selects the attribute producing the largest Information Gain.

Advantage: Simple and effective.

Limitation: Information Gain can favour attributes with many distinct values.

C4.5 — Gain Ratio

C4.5 addresses the attribute-selection limitation of ID3 by using Gain Ratio.

Advantage: Reduces the bias toward attributes with many possible values.

Limitation: The additional calculation makes the process more complex.

CART — Gini Index

CART commonly uses Gini impurity for classification.

Advantage: Computationally convenient and effective for binary classification.

Limitation: The binary-tree structure can sometimes produce deeper trees than a multiway tree.

6.3 Pruning Strategies

Pruning is important because an excessively large Decision Tree can memorize training data and perform poorly on unseen data.

ID3

The original ID3 does not have a dedicated post-pruning mechanism.

This is one of the major limitations of the original algorithm.

C4.5

C4.5 applies **error-based post-pruning** after tree construction. A subtree can be replaced by a leaf when doing so does not significantly reduce classification performance.

CART

CART uses **cost-complexity pruning**, which explicitly considers a trade-off between classification error and tree complexity.

Therefore:

ID3 < C4.5/CART

in terms of the sophistication of built-in pruning mechanisms.

6.4 Interpretability

All three algorithms have a major advantage over many black-box models: they can represent decisions as a sequence of understandable rules.

For example:

```
IF CreditScore > 700
AND RepaymentHistory = Good
THEN Loan = Eligible
```

This makes Decision Trees particularly useful in applications where users need to understand why a prediction was made.

C4.5 can additionally transform trees into rule sets, which further supports understandable classification models.

6.5 Computational Complexity

The computational behaviour of the algorithms depends strongly on:

- Number of records
- Number of attributes
- Number of attribute values
- Continuous vs categorical features
- Missing values
- Tree depth
- Pruning procedure

A recent comparative study examining ID3, C4.5 and CART reported significant differences in tree construction time across datasets. In that study, ID3 and CART were generally faster than C4.5 for the tested datasets, although the exact result depended on the dataset.

This demonstrates an important point:

There is no universal computational winner across all datasets.

6.6 Classification Performance

Classification performance also varies depending on the dataset.

A study comparing multiple Decision Tree strategies on UCI datasets evaluated:

- Classification accuracy
- Tree depth
- Number of leaf nodes
- Tree construction time

The study found differences between ID3, C4.5 and CART across datasets, and reported that ID3 and CART were favourable in several of the evaluated efficiency measures.

Another comparative study examining ID3, C4.5 and CART highlights that the three algorithms can produce different classification performance depending on the characteristics of the dataset.

Therefore, algorithm selection should consider more than accuracy alone.

7. Research Evidence Comparison

Study	Algorithms / Focus	Evaluation	Key Finding
Quinlan (1986)	ID3	Decision-tree induction	Established Information Gain-based tree induction and discussed noise/incomplete data.
Quinlan (1993)	C4.5	Tree construction and practical ML	Extended ID3 with support for numeric/missing data and pruning.
Breiman et al. (1984)	CART	Tree classification/regression	Established binary tree methodology and cost-complexity pruning.
Loh (2010)	CART, C4.5 and related tree methods	Review/comparison	Discussed similarities and differences among tree-structured classifiers.
Liu et al. (2022)	ID3, C4.5, CART and others	Accuracy, runtime, tree height	Demonstrated that performance varies across datasets and evaluation measures.
Aaboub et al. (2023)	ID3, C4.5, CART and newer criteria	Accuracy, depth, leaves, time	Found substantial differences in tree construction time and overall effectiveness across datasets.
Blanc et al. (2023)	Generalized tree learning	Accuracy/scalability	Showed that considering multiple candidate splits rather than only the greedy best split can improve performance while remaining more scalable than optimal-tree approaches.

8. Critical Analysis

Strengths of ID3

- Simple algorithm.
- Easy to understand.
- Information-theoretic basis.
- Suitable for categorical classification.
- Relatively straightforward implementation.

Weaknesses of ID3

- Original method lacks sophisticated pruning.
 - Sensitive to noisy data.
 - Information Gain can favour high-cardinality attributes.
 - Limited handling of continuous features in its original form.
-

Strengths of C4.5

- Improved version of ID3.
- Uses Gain Ratio.
- Supports continuous attributes.
- Handles missing values.
- Includes pruning.
- Produces interpretable models.

Weaknesses of C4.5

- More computational processing than basic ID3.
 - Gain Ratio can have its own selection behaviour depending on the dataset.
 - Large trees can still occur for complex datasets.
-

Strengths of CART

- Binary tree structure.
- Gini criterion is computationally practical.
- Supports both classification and regression.
- Strong pruning framework.
- Can incorporate cost-sensitive considerations.

Weaknesses of CART

- Binary splitting may create deeper trees.
 - Continuous-feature split evaluation can increase computational requirements.
 - Performance can depend strongly on hyperparameters and pruning choices.
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9. Research Gap

The literature demonstrates that ID3, C4.5 and CART have different advantages, but there is **no single attribute-selection criterion that consistently provides the best combination of accuracy, computational efficiency, tree size and interpretability across all datasets.**

Existing comparative studies often evaluate algorithms using fixed criteria and datasets. For example, studies compare accuracy, tree depth, leaf-node count and construction time, but the optimal method can vary according to dataset characteristics.

A further limitation is the **greedy nature of classical Decision Tree learning**. Standard methods typically select the best split at each node rather than considering several promising alternatives. Recent research has shown that considering the top-k candidate attributes can improve predictive performance over standard greedy tree learning while remaining more scalable than fully optimal Decision Tree approaches.

Identified gap

There is an opportunity for an adaptive Decision Tree framework that dynamically selects the splitting criterion and pruning strategy based on dataset characteristics instead of applying a single fixed strategy throughout the tree.

10. Proposed Research Direction

Adaptive Hybrid Decision Tree

A possible improvement is an **Adaptive Hybrid Decision Tree**.

Instead of always using one criterion:

ID3 → Information Gain
C4.5 → Gain Ratio
CART → Gini

the proposed approach would analyse the characteristics of the current dataset/node and select the most appropriate criterion.

Proposed architecture

```
Dataset
↓
Dataset Characteristic Analysis
↓
Evaluate Candidate Splitting Criteria
↓
Information Gain / Gain Ratio / Gini
↓
Select Best Criterion
↓
Generate Candidate Splits
↓
Tree Construction
↓
Validation-Based Pruning
↓
```

Possible decision factors

The system could consider:

- Number of attributes
- Number of categorical attributes
- Number of continuous attributes
- Attribute cardinality
- Class imbalance
- Dataset size
- Noise level
- Missing-value percentage

Expected benefits

1. Better adaptability.
2. Reduced attribute-selection bias.
3. Improved classification performance.
4. Better control of tree complexity.
5. Potentially improved generalization.
6. Retention of Decision Tree interpretability.

This direction is also consistent with recent work questioning purely greedy split selection and investigating broader candidate choices.

11. Discussion

The review indicates that ID3, C4.5 and CART should not simply be ranked as "old," "better," and "best." Each algorithm was designed with different priorities.

ID3 emphasizes simplicity and information-theoretic selection.

C4.5 extends this approach by addressing important practical issues such as continuous attributes, missing values and pruning.

CART provides a more general statistical framework with binary splitting, Gini-based classification and cost-complexity pruning.

Research evidence shows that algorithm performance depends on the characteristics of the dataset. A method that performs well in terms of classification accuracy may not be optimal in terms of runtime or tree complexity.

Therefore, future Decision Tree research should move beyond simple algorithm-versus-algorithm comparisons and investigate **adaptive strategies** that select appropriate splitting and pruning mechanisms according to the data.

12. Conclusion

This research review examined the three classical Decision Tree algorithms **ID3**, **C4.5** and **CART**.

ID3 uses **Information Gain**, while C4.5 improves attribute selection through **Gain Ratio** and introduces stronger support for continuous data, missing values and pruning. CART commonly uses the **Gini Index**, produces binary trees and applies **cost-complexity pruning**.

The comparison shows that:

- **ID3** is simple and interpretable.
- **C4.5** provides a more flexible extension of ID3.
- **CART** provides a strong framework for classification and regression.
- No algorithm consistently dominates across accuracy, runtime, tree complexity and scalability.
- Dataset characteristics strongly influence the appropriate algorithm.

The major research gap identified is the reliance on relatively fixed splitting and pruning strategies. An **Adaptive Hybrid Decision Tree** that dynamically chooses among Information Gain, Gain Ratio and Gini-based strategies and applies validation-driven pruning could provide a promising research direction.

13. References

1. Quinlan, J. R. (1986). **Induction of Decision Trees**. *Machine Learning*, 1, 81–106.
2. Quinlan, J. R. (1993). **C4.5: Programs for Machine Learning**. Morgan Kaufmann.
3. Breiman, L., Friedman, J. H., Olshen, R. A., & Stone, C. J. (1984). **Classification and Regression Trees**. Wadsworth.
4. Loh, W. Y. (2010). **Tree-structured classifiers**. *WIREs Computational Statistics*, 2(3), 364–369.
5. Liu, et al. (2022). **Global Economic Market Forecast and Decision System for IoT and Machine Learning**. *Mobile Information Systems*. The study includes comparative experiments involving ID3, C4.5 and CART.
6. Aaboub, F., Chamlal, H., & Ouaderhman, T. (2023). **Statistical analysis of various splitting criteria for decision trees**. *Proceedings of the Institution of Mechanical Engineers, Part I*.
7. Blanc, G., Lange, J., Pabbaraju, C., Sullivan, C., Tan, L.-Y., & Tiwari, M. (2023). **Harnessing the Power of Choices in Decision Tree Learning**.
8. Hssina, B., Merbouha, A., Ezzikouri, H., & Erritali, M. (2014). **A comparative study of decision tree ID3 and C4.5**. *International Journal of Advanced Computer Science and Applications*, 4(2).

Structure of the Report:

S.No.	Paper Section	Expected Content
1	Title	Specific and concise title related to the selected research question/topic.
2	Abstract	Brief summary of the research problem, review methodology, major findings, research gap, and conclusion.
3	Keywords	4–6 important keywords related to the selected topic.
4	1. Introduction	Introduce the AI topic, background, importance, motivation, and scope of the review.
5	2. Research Problem and Objectives	Clearly state the research problem and specific objectives of the review.
6	3. Literature Review	Review relevant research papers and summarize major approaches, methodologies, findings, and contributions.
7	4. Comparative Analysis	Compare the reviewed approaches based on suitable parameters such as accuracy, complexity, interpretability, computational cost, scalability, and generalization.
8	5. Critical Analysis	Discuss strengths, weaknesses, limitations, assumptions, and practical challenges identified across the reviewed studies.
9	6. Research Gap	Identify unresolved problems, limitations in existing research, and areas requiring further investigation.
10	7. Proposed Research Direction	Suggest possible solutions, improvements, or future research directions based on the identified gap.
11	8. Discussion	Discuss the overall insights obtained from the review and their implications for AI research and applications.

12	9. Conclusion	Summarize the major findings, key insights, research gap, and future direction.
13	References	List all research papers, books, datasets, websites, and other sources using a consistent citation style.

Evaluation Rubrics:

Criterion	Weightage (%)	Excellent	Good	Average	Poor
1. Research Problem Identification and Understanding	10%	9–10% – Clearly identifies the research problem, objectives, scope, and relevance to AI learning approaches.	7–8% – Research problem and objectives are clearly identified with minor omissions.	5–6% – Basic understanding of the research problem with some gaps.	0–4% – Problem is unclear, inappropriate, or poorly understood.
2. Literature Search and Source Selection	10%	9–10% – Uses diverse, highly relevant, recent, and credible research papers, journals, conferences, and technical sources.	7–8% – Uses relevant and credible sources with minor limitations.	5–6% – Uses a limited number of relevant sources.	0–4% – Sources are insufficient, unreliable, or largely irrelevant.

3. Literature Review and Synthesis	15%	13–15% – Provides a comprehensive review and synthesizes findings across multiple studies rather than merely summarizing them.	10–12% – Good review with meaningful comparison of major studies.	7–9% – Basic review with limited synthesis and comparison.	0–6% – Superficial review with little understanding of existing research.
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4. Analysis of AI Techniques / Methods	15%	13–15% – Provides technically accurate and in-depth analysis of relevant methods such as decision trees, statistical learning, neural networks, kernel methods, or reinforcement learning.	10–12% – Provides good technical analysis with minor gaps.	7–9% – Provides basic technical description with limited analysis.	0–6% – Methods are poorly understood or inaccurately described.
5. Comparative Analysis of Research Approaches	10%	9–10% – Critically compares approaches based on accuracy, complexity, interpretability, scalability, computational cost, and application suitability.	7–8% – Provides meaningful comparison using several parameters.	5–6% – Basic comparison with limited criteria.	0–4% – Little or no meaningful comparison.

6. Critical Evaluation of Research Findings	10%	9–10% – Critically evaluates strengths, weaknesses, assumptions, experimental methods, and findings of existing studies.	7–8% – Provides good critical evaluation with minor gaps.	5–6% – Some critical observations are provided but analysis is limited.	0–4% – Mostly descriptive with little critical evaluation.
7. Identification of Research Gaps	10%	9–10% – Clearly identifies significant research gaps, limitations, unresolved issues, and opportunities for further investigation.	7–8% – Identifies relevant research gaps with reasonable justification.	5–6% – Identifies basic gaps but provides limited justification.	0–4% – Research gaps are missing or incorrectly identified.
8. Research Insights and Recommendations	5%	5% – Provides insightful conclusions and technically feasible recommendations for future research or applications.	4% – Provides relevant conclusions and recommendations.	2–3% – Basic conclusions with limited recommendations.	0–1% – Conclusions or recommendations are absent or inappropriate.
9. Technical Report and Referencing	10%	9–10% – Report is well structured, technically accurate, professionally written, and uses a consistent citation/reference style.	7–8% – Well-structured report with minor writing or referencing issues.	5–6% – Adequate report but contains several structural or referencing issues.	0–4% – Poorly structured, inadequately referenced, or incomplete report.

10. Originality, Academic Integrity and Use of Sources	5%	5% – Demonstrates original synthesis, proper citation, and strong academic integrity with no inappropriate copying.	4% – Good originality and proper citation with minor issues.	2–3% – Some original analysis but significant dependence on sources.	0–1% – Evidence of substantial copying, poor citation, or academic- integrity concerns.
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